



Seat No : _____

No. of Printed Pages: 1

[3]

SARDAR PATEL UNIVERSITY

BCA Semester-2 External Examination – April-2026 [NEP]

US02IDBCA05 || Digital Electronics

Date : 16-04 -2026

Time : 10.00 a.m. – 11.30 a.m.

Total : 25 Marks

Q-1 Select the appropriate option for the following questions: **[05]**

1) De Morgan's first theorem says that a NOR gate is equivalent to a _____

- | | |
|----------------|-----------------|
| A. Bubbled OR | B. Bubbled AND |
| C. Bubbled XOR | D. Bubbled NAND |

2) An invert gate is also called a _____ gate.

- | | |
|---------|---------|
| A. NOR | B. NOT |
| C. XNOR | D. NAND |

3) In half adder AND gate's output is _____

- | | |
|--------------|------------------|
| A. Carry | B. Sum |
| C. Remainder | D. None of these |

4) The basic building block of sequential logic circuit is _____

- | | |
|--------------|-------------|
| A. Flip-Flop | B. OR Gate |
| C. NAND Gate | D. AND Gate |

5) The truth table for an R-S Flip-Flop has how many VALID entries?

- | | |
|------|------|
| A. 3 | B. 2 |
| C. 1 | D. 4 |

Q-2 Explain Encoder with circuit and truth table. **[10]**

OR

Q-2 Define logic gate. Draw electronic symbol and explain AND, OR, NOT gates with truth table. **[10]**

Q-3 Explain D Flip-Flop (Clocked & Unlocked) with diagrams. **[10]**

OR

Q-3 Explain Left-Shift and Right-Shift register with diagram. **[10]**